Electrochemical and Chemical Coatings

Eka Nurfani

Department of Materials Engineering, Institut Teknologi Sumatera (Itera)
E-mail: eka.nurfani@mt.itera.ac.id

February 13, 2025

ITERA

Outline

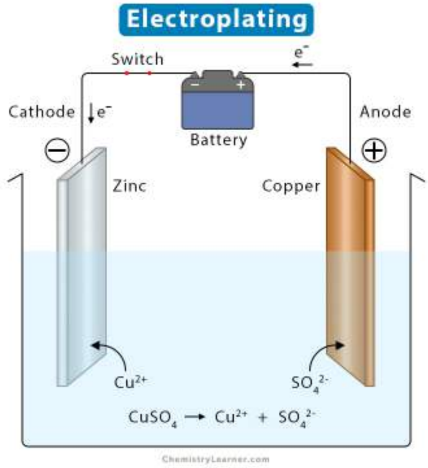
Outline

Electrochemical Coating

- Coating onto the surface via electrochemical reactions
- The coating can be (a) metallic, (b) metal oxide or (c) conductive polymer.
- Metallic coating: **Electroplating**
- Metal oxide, conductive polymer: **Anodizing**
- **Electroless deposition**

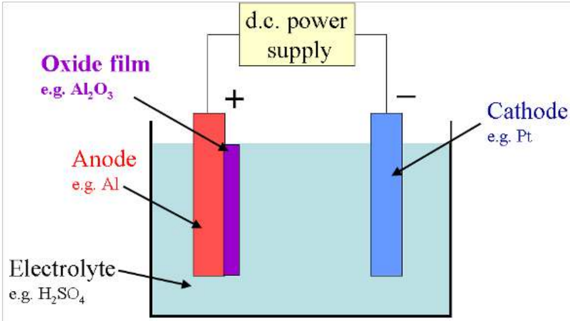
Electroplating of Copper

- An electro-chemical reaction
- Cathode: Metals/alloys coating
- Anode: Metal oxides
- Conductive solution: ionic species
- Transfer of electrons



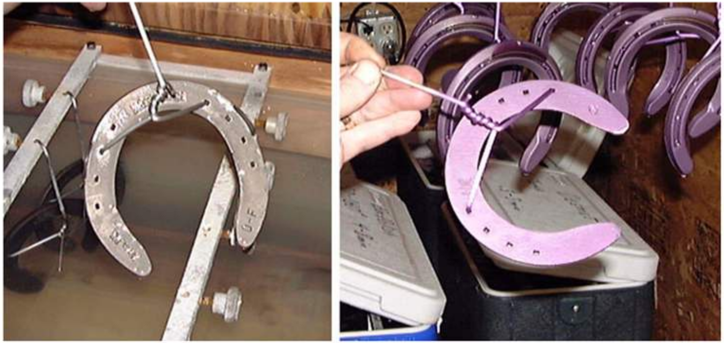
Anodizing

- An electrolytic passivation process.
- To form a thick oxide layer on a metal.
- Metal oxide forms on the anode.



$$2Al + 3H_2O - 6e^- = Al_2O_3 + 6H^2 \tag{1}$$

Example of anodizing



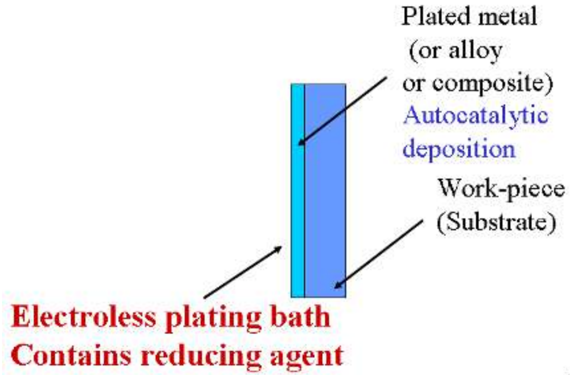
Before After anodizing

Function of anodized surface

1. Corrosion resistance, thick oxide layer.
2. Aesthetic surface finish.

Electroless Deposition

- Electroplating: consisting of two electrodes, electrolyte, and external current source.
- Electroless deposition: this process uses only one electrode and no external source of electric current.
- The solution needs to contain a reducing agent.



Definition: Electron transfer reactions

- Oxidizing agent + n e- = Reducing agent
- Oxidizing agents get reduced
- Reducing agents get oxidized
- Oxidation is a loss of electrons
- Reduction is a gain of electrons

Outline

Typical steps in the electroplating of metals

- Cleaning with organic solvent or aqueous alkaline; to remove dirt or grease.
- Is oxides cover the surface as a result of corrosion, clean with acid.
- Rinse with water to neutralize the surface.
- Electroplate metals under controlled conditions.
- Rinse with water and dry.
- Additional step: heat treatment in air or vacuum environment

What is the Job of the Bath?

- Provides an electrolyte: to conduct electricity ionically
- Provides a source of the metal to be plated: as dissolved metal salts leading to metal ions
- Allows the anode reaction to take place: usually metal dissolution or oxygen evolution
- Wets the cathode work-piece: allowing good adhesion to take place
- Helps to stabilise temperature: acts as a heating/cooling bath

What is in a Bath?

- Ions of the metal to be plated, e.g., Ni2+ (nickel ions) added mostly as the sulfate
- Conductive electrolyte: NiSO4, H3BO4, NiCl2
- Nickel anode dissolution promoter: mempercepat atau meningkatkan laju pelarutan nikel dari anoda selama proses elektrokimia
- pH buffer stops cathode getting too alkaline: Boric acid (H3BO3)

Current efficiency

- pH changes accompany electrode reactions wherever H^+ or OH^- ions are involved.
- In acid, hydrogen evolution occurs on the surface of the cathode. This will result in a localized increase in pH near the surface of the electrode. $2H^+ + 2e^- \rightarrow H_2$
- In acid, oxygen evolution occurs on the surface of the anode. This will result in a drop of pH near the surface of the electrode. $H_2O - 2e^- \rightarrow 2H^+ + 0.5O_2$

Hydrogen evolution adalah proses pembentukan gas hidrogen (H2) sebagai hasil dari reaksi elektrokimia, biasanya terjadi pada katoda dalam elektrolisis atau dalam reaksi korosi tertentu.

Current efficiency

Current efficiency adalah perbandingan antara jumlah aktual endapan logam, M_{actual} dengan yang dihitung secara teoritis dari Hukum Faraday, M_{theory}.

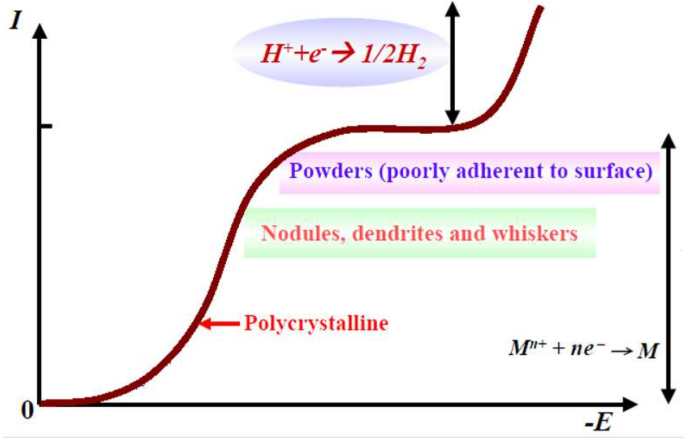
$$I_{eff} = \frac{M_{actual}}{M_{theory}} \times 100\% \tag{2}$$

Parameters

Parameters that may influence the quality of electrodeposits:

- Current density (low to high current)
- The nature of anions/cations in the solution
- Bath composition, temperature, fluid flow
- Type of current waveform
- the presence of impurities
- physical and chemical nature of the substrate surface

An example of Current vs. Potential Curve for electroplating of metal



Watts Nickel

Watts Nickel adalah larutan elektrodeposisi nikel yang paling umum digunakan dalam proses electroplating (pelapisan elektrokimia).

Larutan ini dikembangkan oleh Oliver P. Watts pada tahun 1916 dan masih digunakan luas hingga saat ini karena menghasilkan lapisan nikel yang seragam, halus, dan tahan korosi.

Typical Recipe and Conditions Watts Nickel

Nickel sulphate: 330 g/L
Nickel chloride: 45 g/L
Boric acid: 40 g/L
Temperature: 60 °C
pH: 4
Current density: 2-10 A dm^{-2}

Faraday's Laws of Electrolysis

Amount of material = amount of electrical energy

$$n = \frac{q}{zF} \tag{3}$$

- n = amount of material (mol)
- q = electrical charge (C)
- z = number of electrons (C)
- F = Faraday constant (mol^{-1})

Expanded Relationship

$$\frac{w}{M} = \frac{It}{zF} \tag{4}$$

$$\begin{aligned} \text{n} &= \text{amount of material} \\ \text{w} &= \text{mass of material} \\ \text{M} &= \text{molar mass of material} \\ \text{I} &= \text{current} \\ \text{t} &= \text{time} \\ \text{z} &= \text{number of electrons} \\ \text{F} &= \text{Faraday constant} \end{aligned}$$

Average thickness

$$x = \frac{MIt}{\rho AzF} \tag{5}$$

$$\begin{aligned} \text{M} &= \text{molar mass of metal} \\ \text{I} &= \text{current} \\ \text{t} &= \text{time} \\ \text{z} &= \text{number of electrons} \\ \text{F} &= \text{Faraday constant} \\ \text{x} &= \text{thickness of plating} \end{aligned}$$

Question - Nickel Plating

Nickel is plated from a Watts bath at a current density of 3 A dm^{-2}. The current efficiency is 96%. The molar mass of nickel is 58.71 g mol^{-1}. The density of nickel is 8.90 g cm^{-3}. The Faraday constant is 96485 C mol^{-1}.
Luasan area coating 10x10 cm.
What will be the average plating thickness in 1 hour?

Answer - Nickel Plating

Assume that the reaction is:

$Ni^{2+} + 2e^{-} = Ni$

So, two electrons are involved for every Ni atom, and z = 2

The current density used in plating nickel is 96% of the total current, i.e., 0.96x3 A dm^{-2}.

The average deposit thickness is given by:

$x = \frac{MIt}{\rho AzF}$

Summary

- Elektrodeposisi merupakan teknik pelapisan yang serbaguna.
- Terdapat kontrol yang tinggi terhadap ketebalan endapan.
- Banyak logam yang dapat dilapis secara elektro dari rendaman air.
- Beberapa paduan, polimer konduktif, dan komposit juga dapat digunakan.
- Laju pelapisan elektro dapat dinyatakan melalui Hukum elektrolisis Faraday.

Outline

Outline

Outline

THANK YOU